Signals and Systems

Lecture 14: Discrete-time Fourier Representation

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Adapted from the materials provided on
the MIT OpenCourseWare

Signal Processing

Modern audio systems process sounds digitally.



DT Fourier Series and Frequency Response

Today: frequency representations for DT signals and systems.

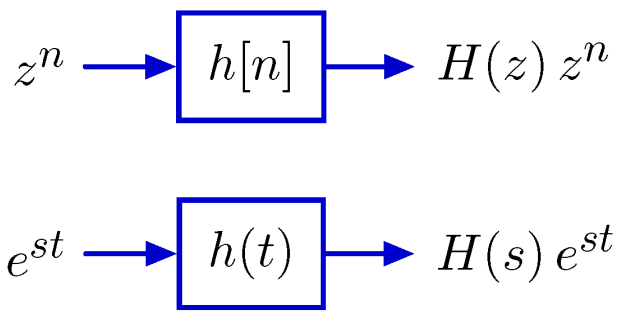
Complex Geometric Sequences

Complex geometric sequences are eigenfunctions of DT LTI systems.

Find response of DT LTI system ($h[n]$) to input $x[n] = z^n$.

$$y[n] = (h * x)[n] = \sum_{k=-\infty}^{\infty} h[k] z^{n-k} = z^n \sum_{k=-\infty}^{\infty} h[k] z^{-k} = H(z)\, z^n \,.$$

Complex geometrics (DT): analogous to complex exponentials (CT)



Rational System Functions

A system described by a linear difference equation with constant coefficients $\rightarrow$ system function that is a ratio of polynomials in z.

Example:

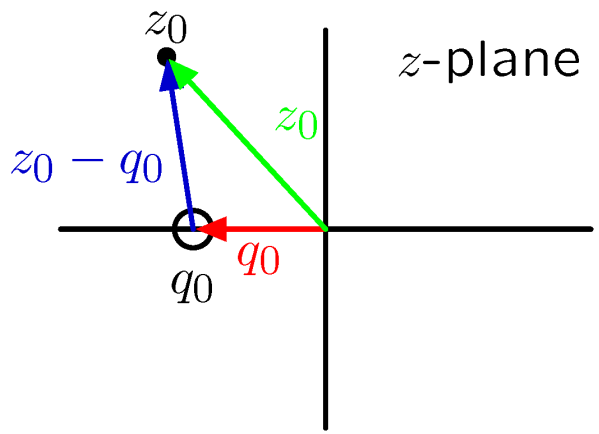
$$y[n-2] + 3y[n-1] + 4y[n] = 2x[n-2] + 7x[n-1] + 8x[n]$$

$$H(z) = \frac{2z^{-2} + 7z^{-1} + 8}{z^{-2} + 3z^{-1} + 4} = \frac{2 + 7z + 8z^2}{1 + 3z + 4z^2} \equiv \frac{N(z)}{D(z)}$$

DT Vector Diagrams

Factor the numerator and denominator of the system function to make poles and zeros explicit.

$$H(z_0) = K\frac{(z_0 - q_0)(z_0 - q_1)(z_0 - q_2)\cdots}{(z_0 - p_0)(z_0 - p_1)(z_0 - p_2)\cdots}$$



Each factor in the numerator/denominator corresponds to a vector from a zero/pole (here q_0) to z_0, the point of interest in the z-plane.

Vector diagrams for DT are similar to those for CT.

DT Vector Diagrams

Value of $H(z)$ at $z = z_0$ can be determined by combining the contributions of the vectors associated with each of the poles and zeros.

$$H(z_0) = K\frac{(z_0 - q_0)(z_0 - q_1)(z_0 - q_2)\cdots}{(z_0 - p_0)(z_0 - p_1)(z_0 - p_2)\cdots}$$

The magnitude is determined by the product of the magnitudes.

$$|H(z_0)| = |K|\,\frac{|(z_0 - q_0)||(z_0 - q_1)||(z_0 - q_2)|\cdots}{|(z_0 - p_0)||(z_0 - p_1)||(z_0 - p_2)|\cdots}$$

The angle is determined by the sum of the angles.

$$\angle H(z_0) = \angle K + \angle(z_0 - q_0) + \angle(z_0 - q_1) + \cdots - \angle(z_0 - p_0) - \angle(z_0 - p_1) - \cdots$$

DT Frequency Response

Response to eternal sinusoids.

Let $x[n] = \cos \Omega_0 n$ (for all time):

$$x[n] = \frac{1}{2}\left(e^{j\Omega_0 n} + e^{-j\Omega_0 n}\right) = \frac{1}{2}\left(z_0^n + z_1^n\right)$$

where $z_0 = e^{j\Omega_0}$ and $z_1 = e^{-j\Omega_0}$.

The response to a sum is the sum of the responses:

$$\begin{aligned} y[n] &= \frac{1}{2}\left(H(z_0)\, z_0^n + H(z_1)\, z_1^n\right) \\ &= \frac{1}{2}\left(H(e^{j\Omega_0})\, e^{j\Omega_0 n} + H(e^{-j\Omega_0})\, e^{-j\Omega_0 n}\right) \end{aligned}$$

Conjugate Symmetry

For physical systems, the complex conjugate of $H(e^{j\Omega})$ is $H(e^{-j\Omega})$.

The system function is the Z transform of the unit-sample response:

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n}$$

where $h[n]$ is a real-valued function of n for physical systems.

$$H(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\Omega n}$$

$$H(e^{-j\Omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{j\Omega n} \equiv \left(H(e^{j\Omega}) \right)^*$$

DT Frequency Response

Response to eternal sinusoids.

Let $x[n] = \cos \Omega_0 n$ (for all time), which can be written as

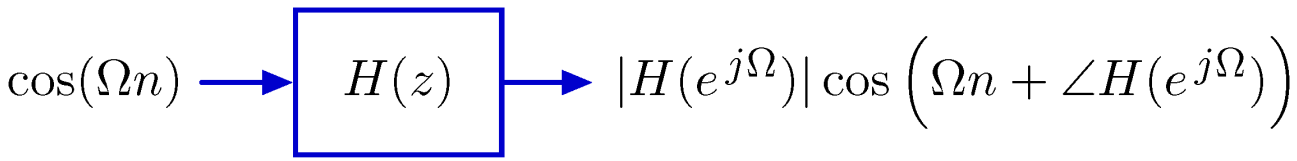
$$x[n] = \frac{1}{2}\left(e^{j\Omega_0 n} + e^{-j\Omega_0 n}\right).$$

Then

$$\begin{aligned} y[n] &= \frac{1}{2}\left(H(e^{j\Omega_0})e^{j\Omega_0 n} + H(e^{-j\Omega_0})e^{-j\Omega_0 n}\right) \\ &= \mathrm{Re}\left\{H(e^{j\Omega_0})e^{j\Omega_0 n}\right\} \\ &= \mathrm{Re}\left\{|H(e^{j\Omega_0})|e^{j\angle H(e^{j\Omega_0})}e^{j\Omega_0 n}\right\} \\ &= |H(e^{j\Omega_0})|\mathrm{Re}\left\{e^{j\Omega_0 n + j\angle H(e^{j\Omega_0})}\right\} \\ y[n] &= \left|H(e^{j\Omega_0})\right| \cos\left(\Omega_0 n + \angle H(e^{j\Omega_0})\right) \end{aligned}$$

Frequency Response

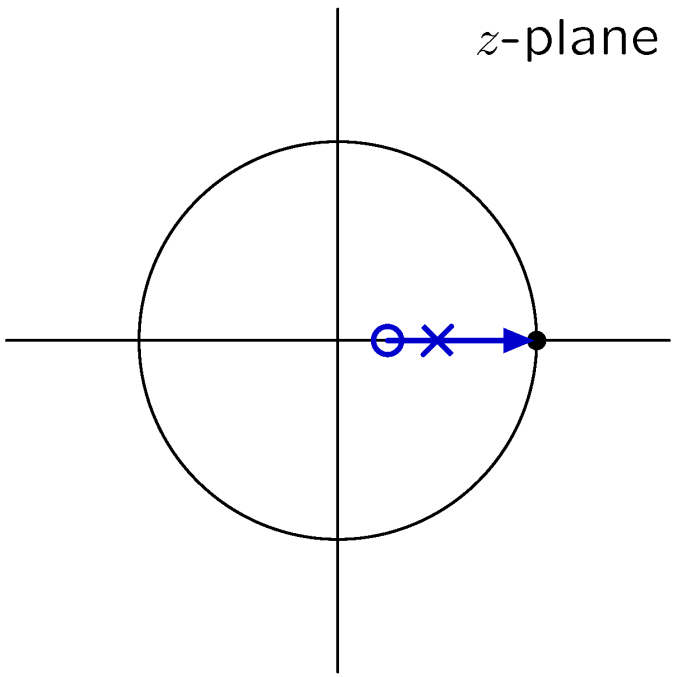
The magnitude and phase of the response of a system to an eternal cosine signal is the magnitude and phase of the system function evaluated on the unit circle.

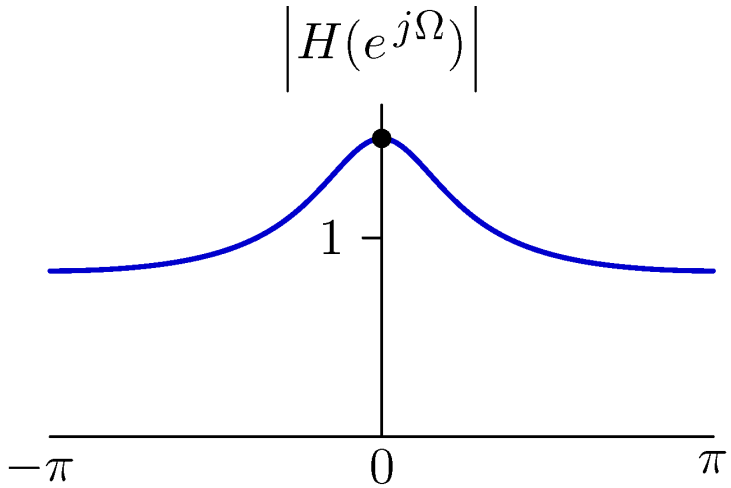


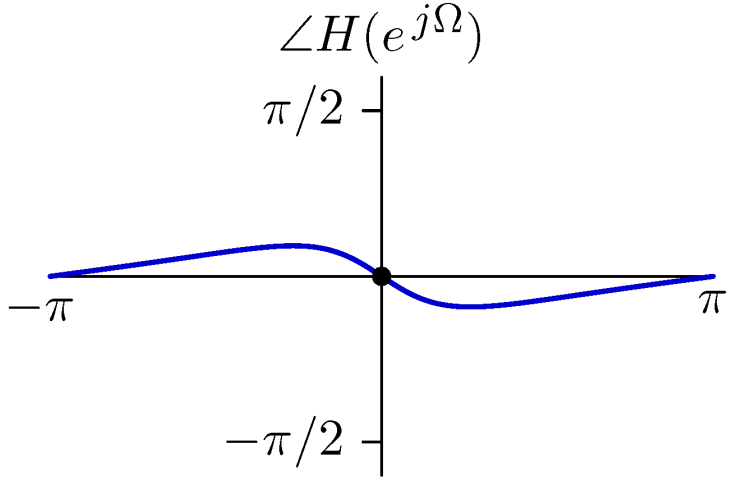
$$H(e^{j\Omega}) = H(z)|_{z=e^{j\Omega}}$$

Finding Frequency Response with Vector Diagrams

$$H(z) = \frac{z - q_1}{z - p_1}$$

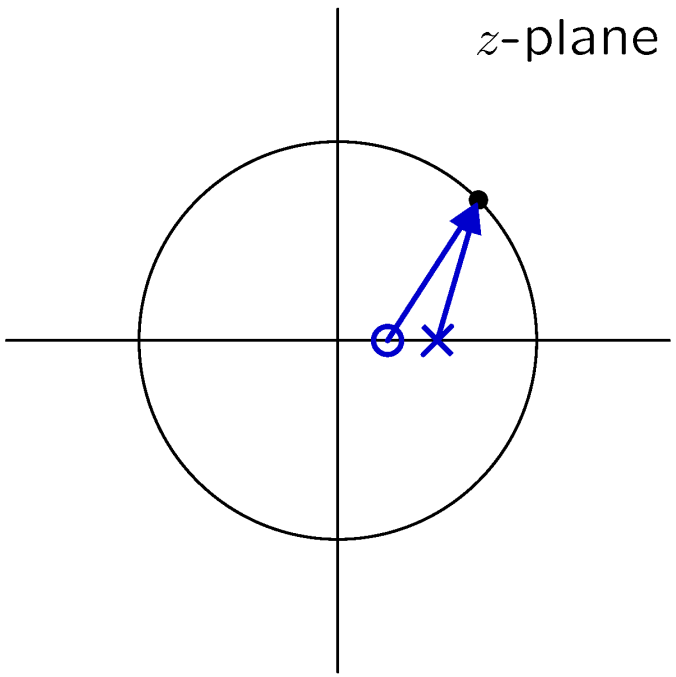


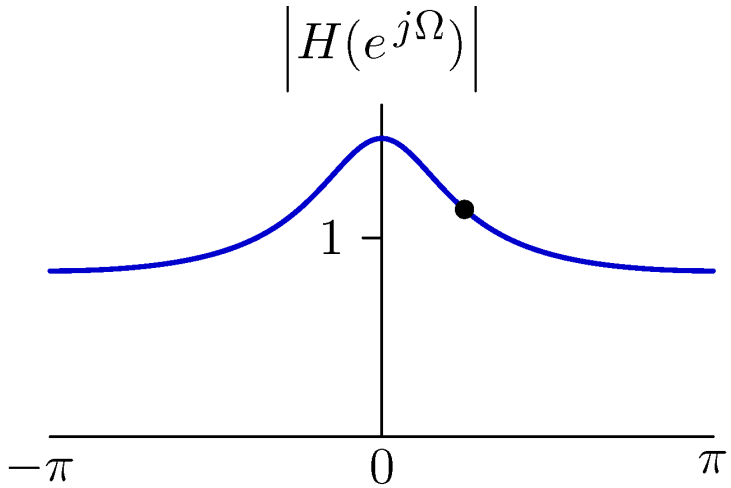


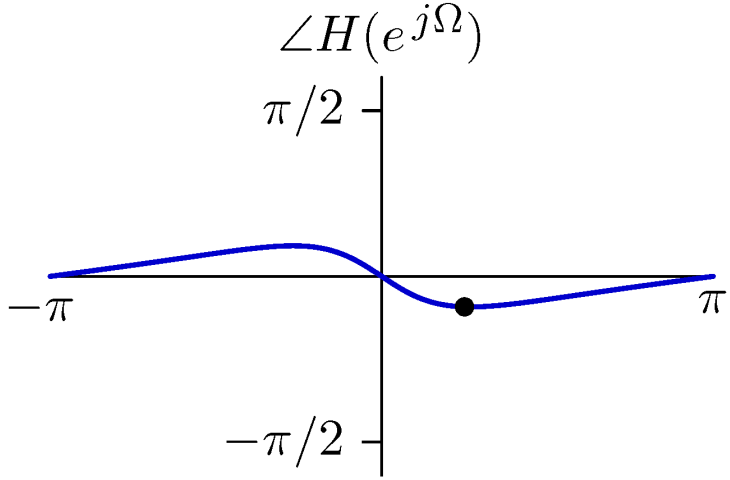


Finding Frequency Response with Vector Diagrams

$$H(z) = \frac{z - q_1}{z - p_1}$$

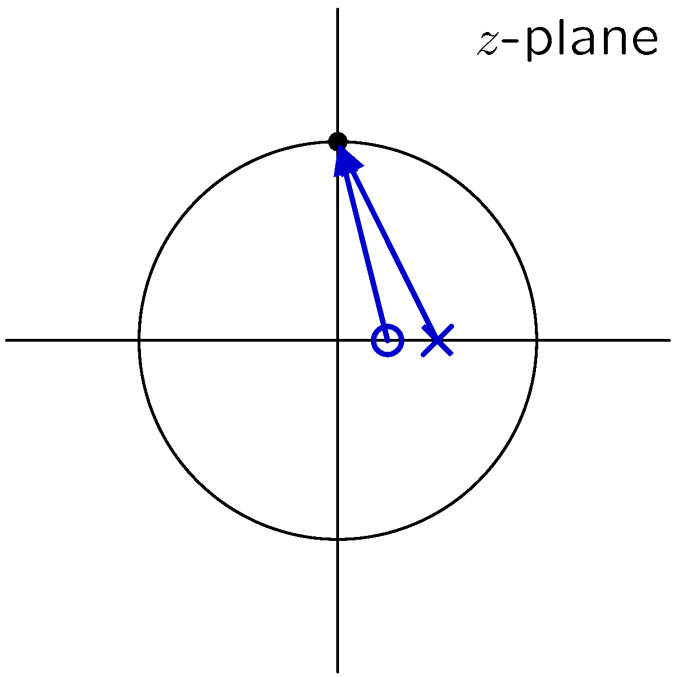


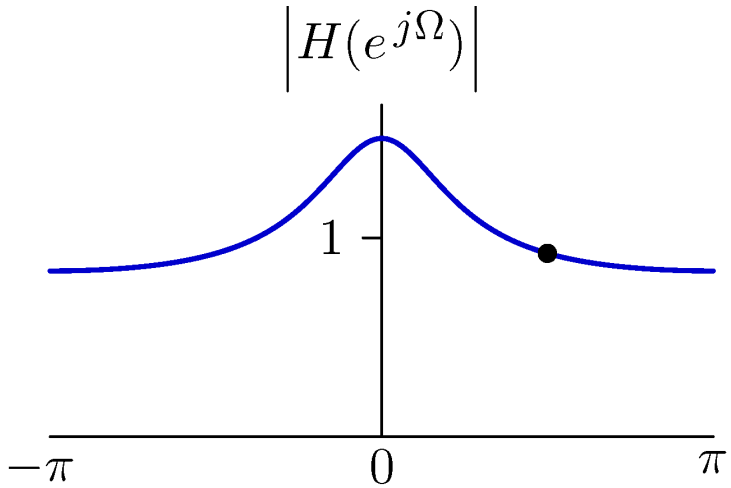


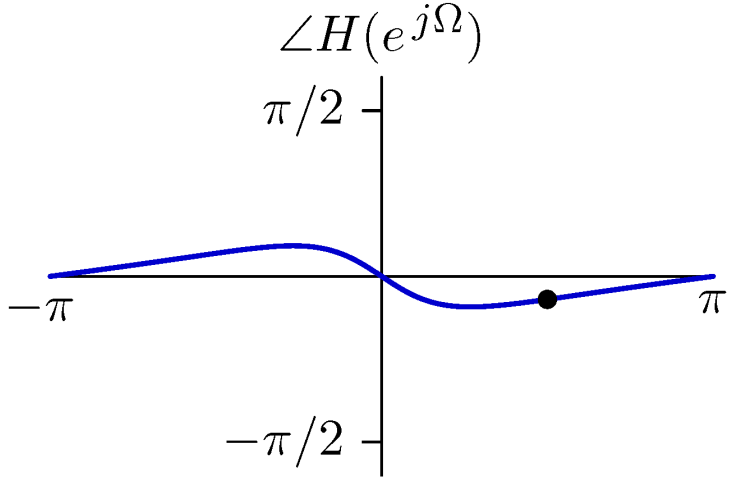


Finding Frequency Response with Vector Diagrams

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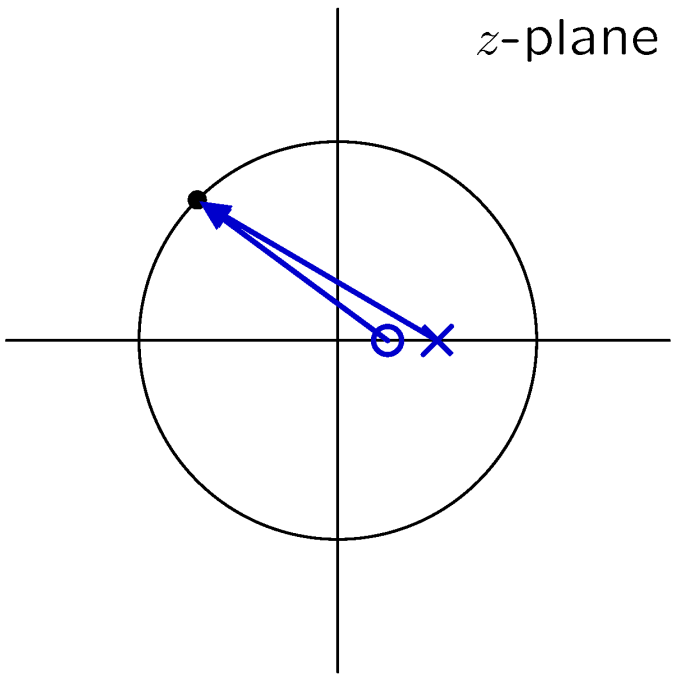


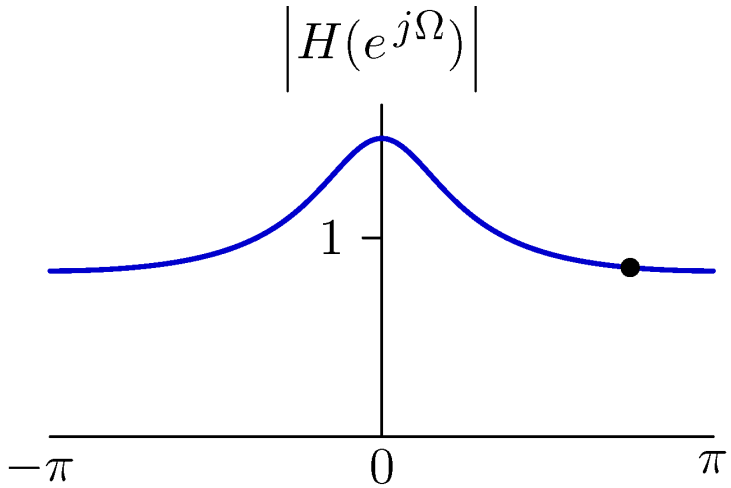


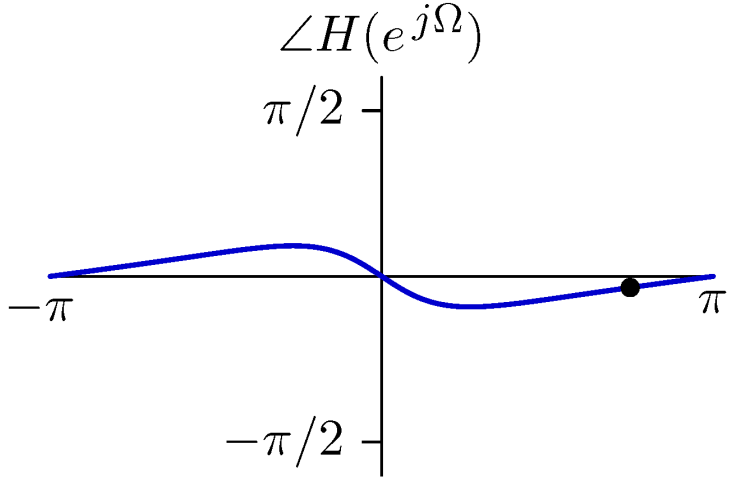


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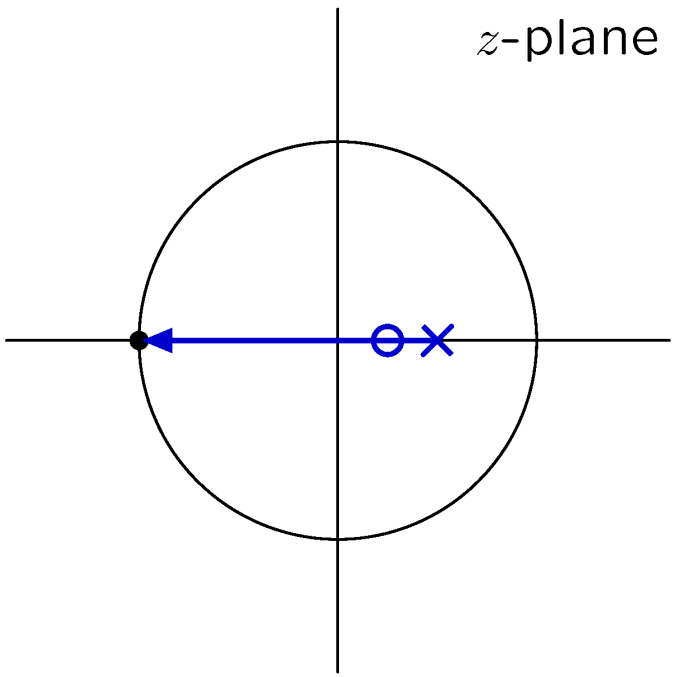


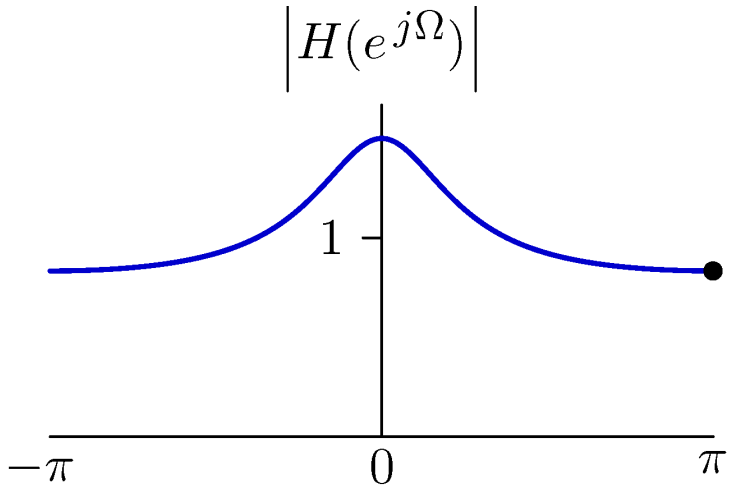


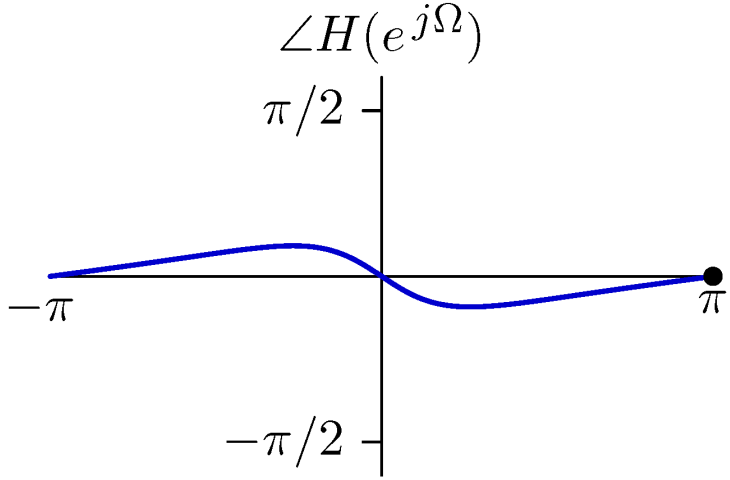


Finding Frequency Response with Vector Diagrams

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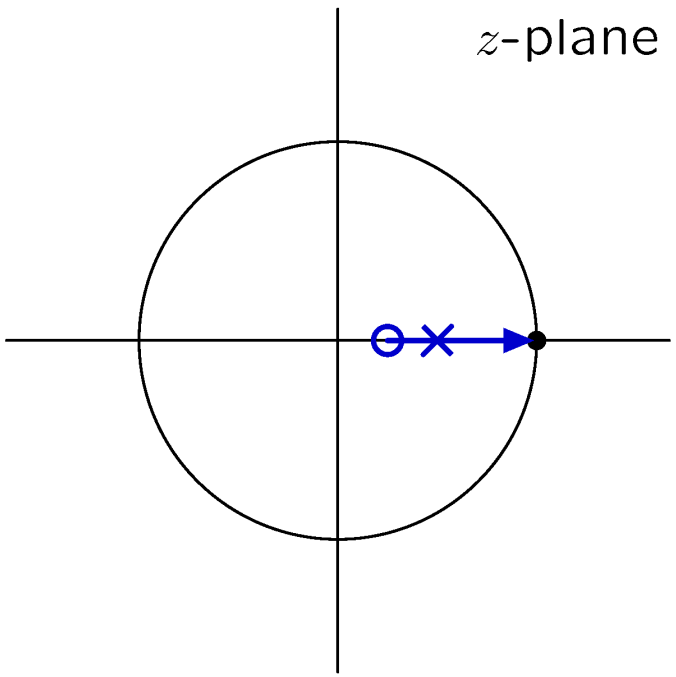


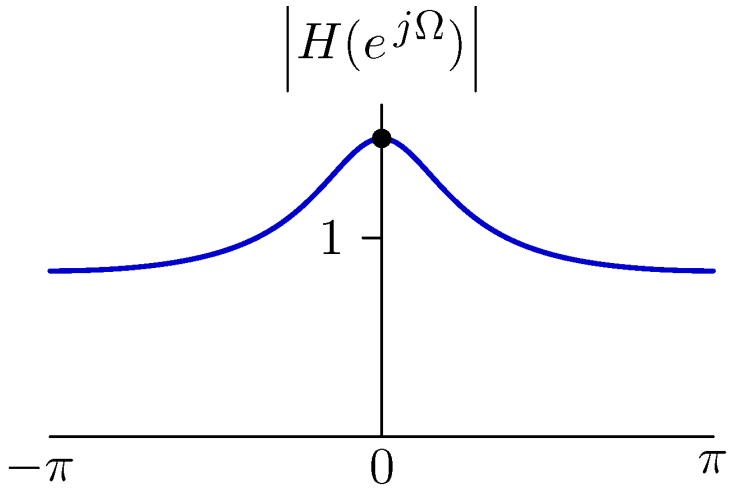


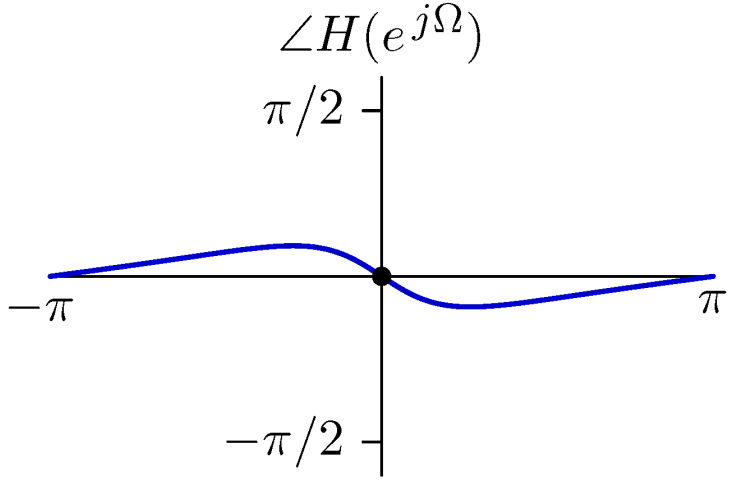


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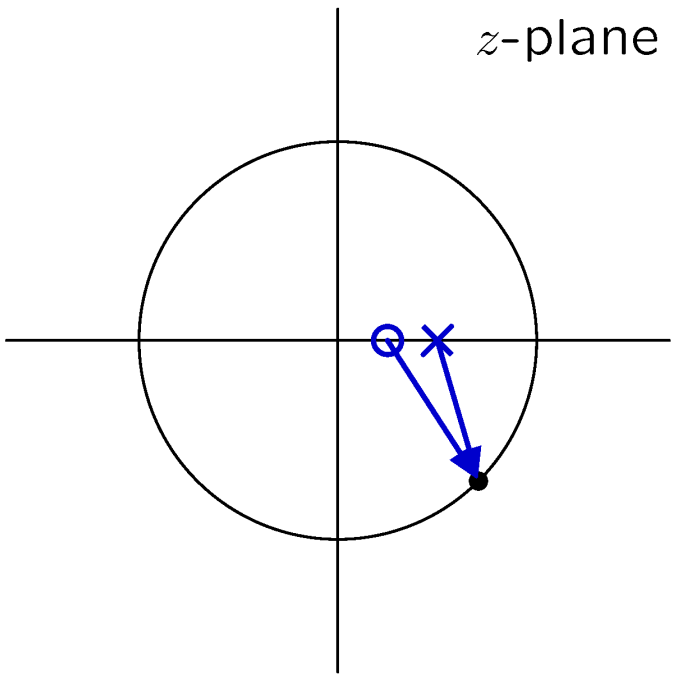


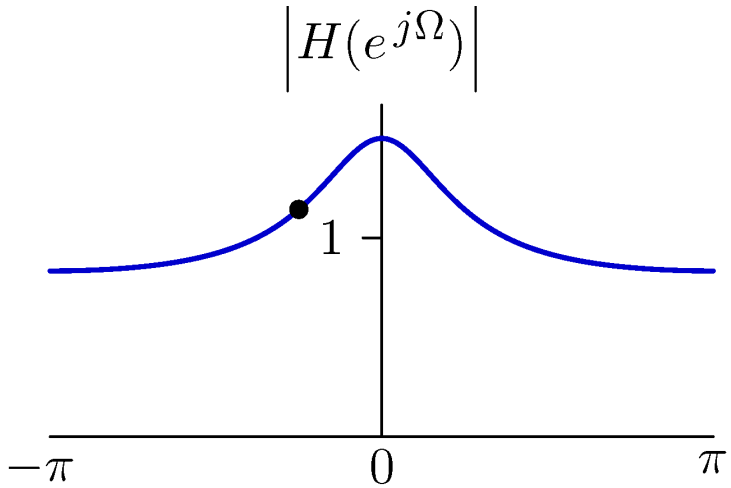


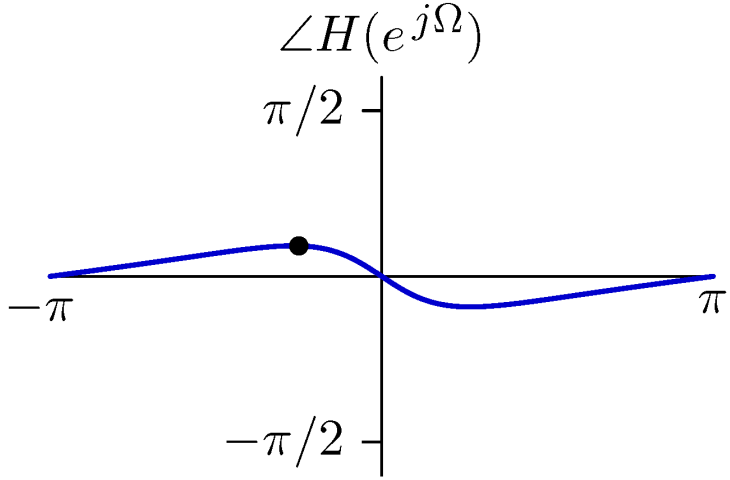


Finding Frequency Response with Vector Diagrams

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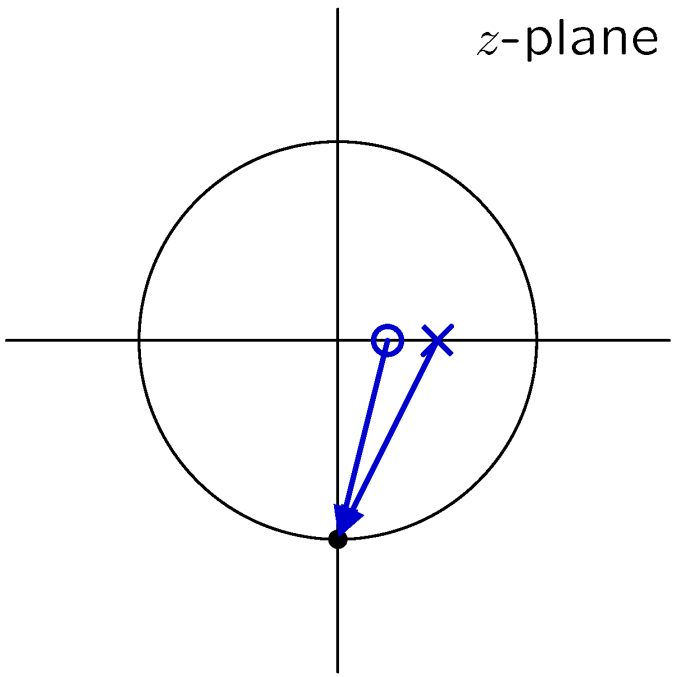


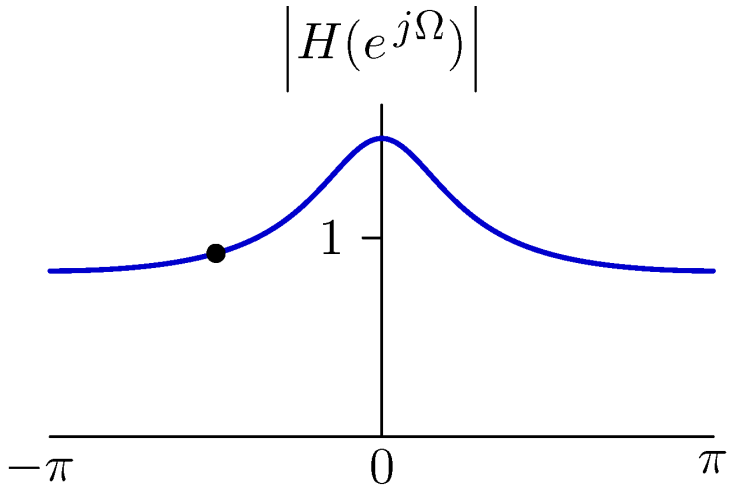


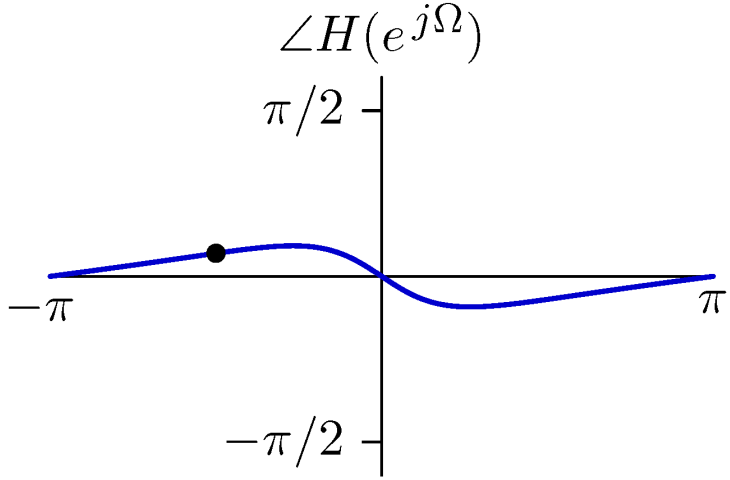


Finding Frequency Response with Vector Diagrams

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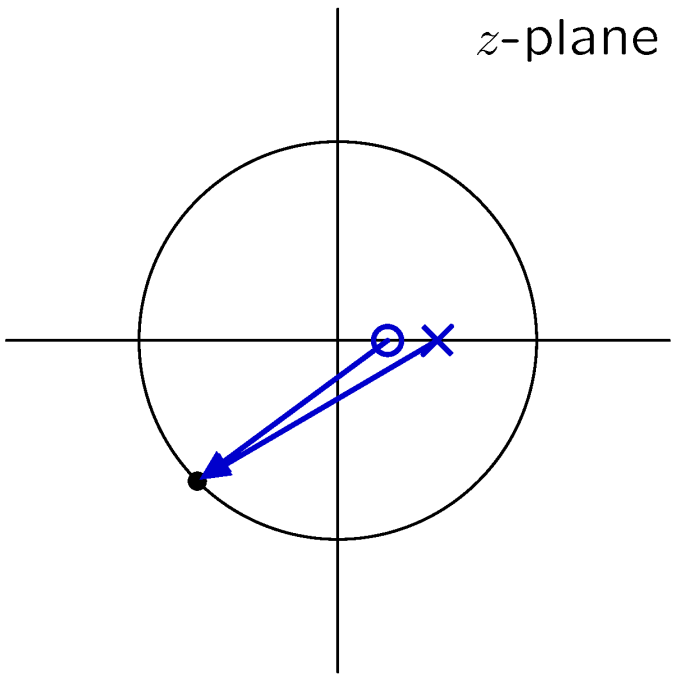


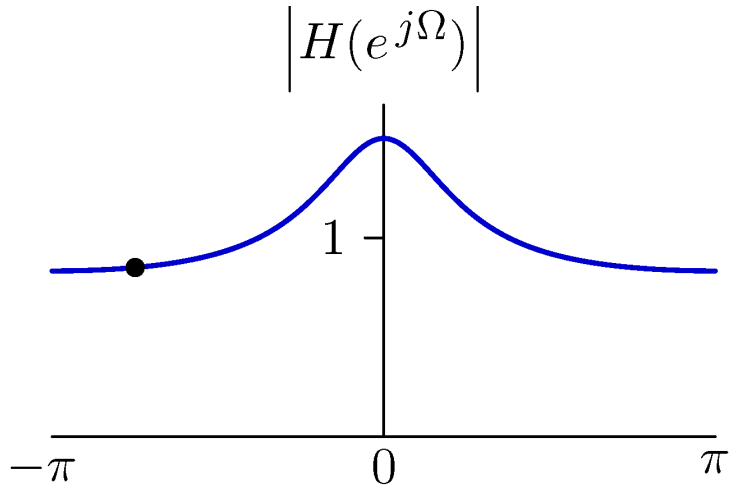


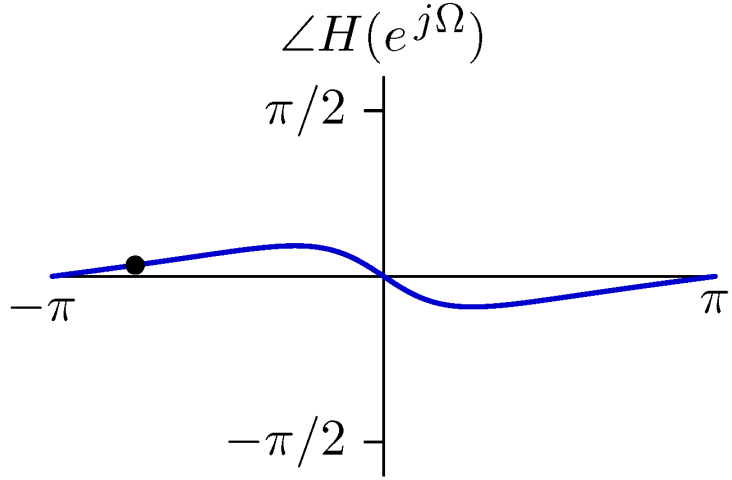


Finding Frequency Response with Vector Diagrams

$$H(z) = \frac{z - q_1}{z - p_1}$$

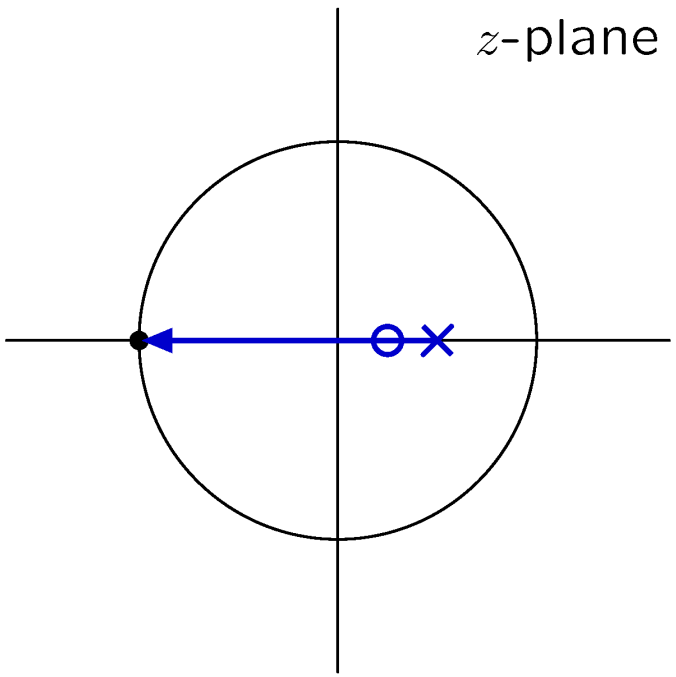


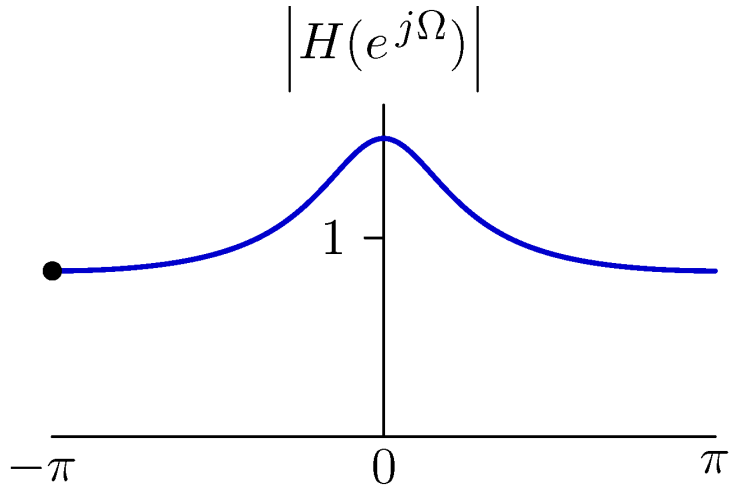


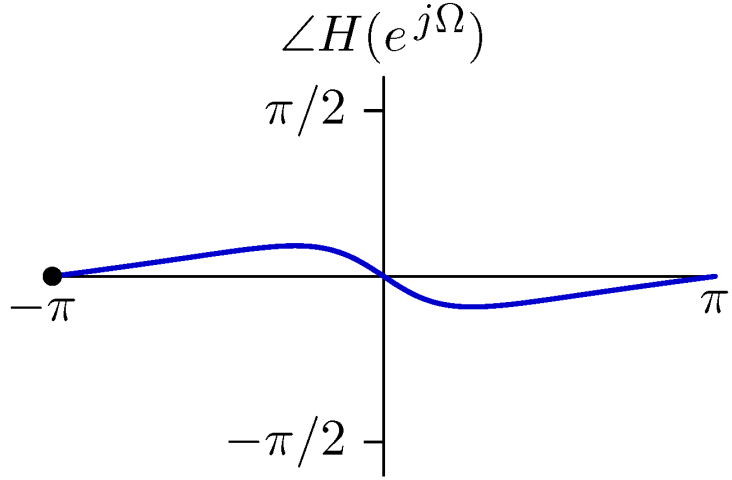


Finding Frequency Response with Vector Diagrams

$$H(z) = \frac{z - q_1}{z - p_1}$$



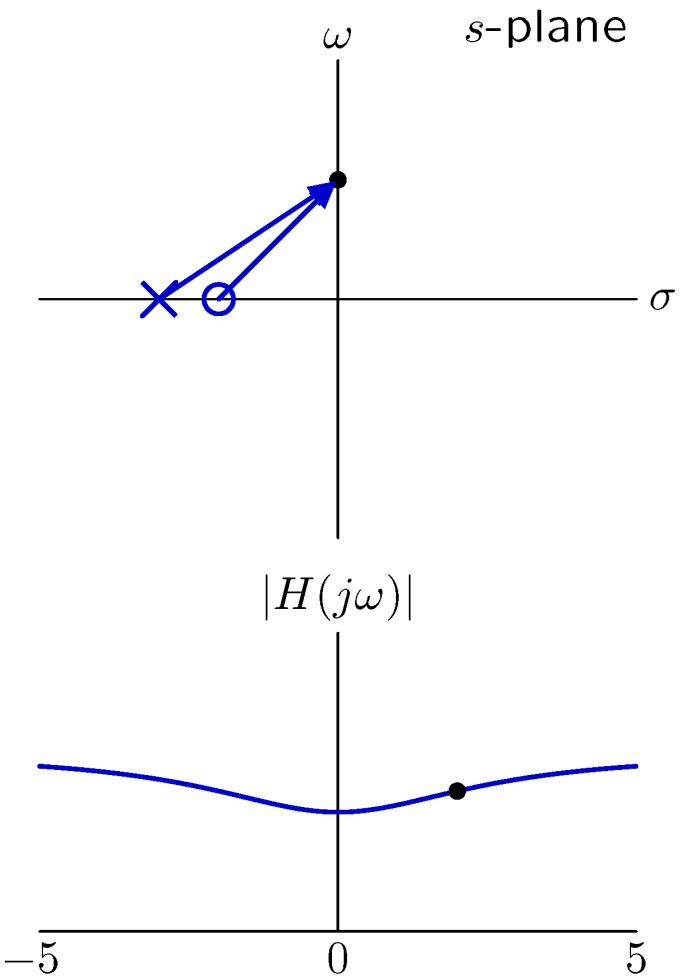


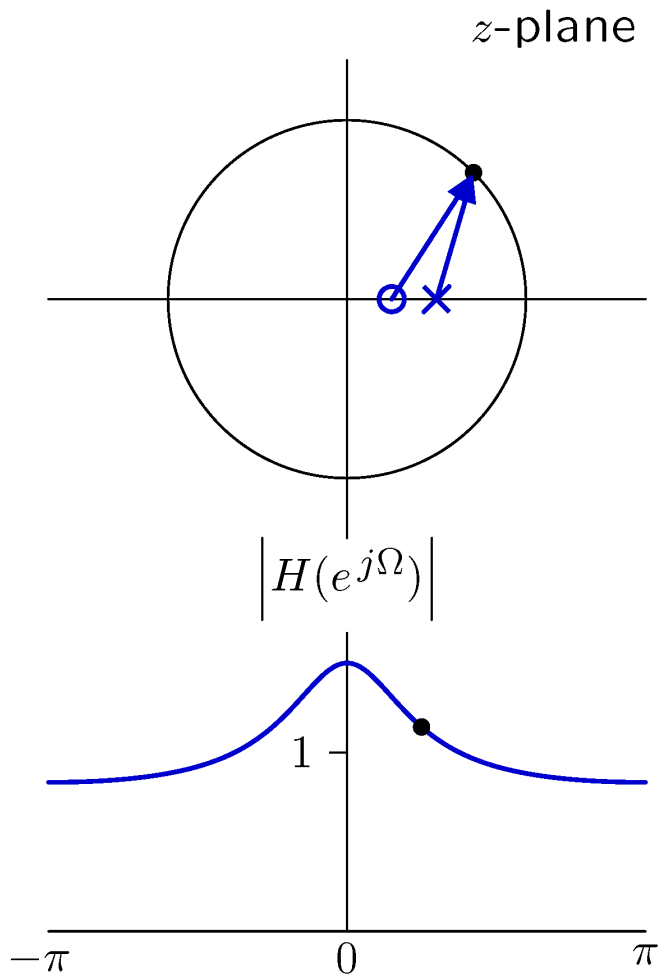


Comparision of CT and DT Frequency Responses

CT frequency response: $H(s)$ on the imaginary axis, i.e., $s = j\omega$.

DT frequency response: $H(z)$ on the unit circle, i.e., $z = e^{j\Omega}$.





Periodicity of DT Frequency Responses

DT frequency responses are periodic functions of Ω, with period 2π.

If $\Omega_2 = \Omega_1 + 2\pi k$ where k is an integer then

$$H(e^{j\Omega_2}) = H(e^{j(\Omega_1+2\pi k)}) = H(e^{j\Omega_1}e^{j2\pi k}) = H(e^{j\Omega_1})$$

The periodicity of $H(e^{j\Omega})$ results because $H(e^{j\Omega})$ is a function of $e^{j\Omega}$, which is itself periodic in Ω. Thus DT complex exponentials have many "aliases."

$$e^{j\Omega_2} = e^{j(\Omega_1+2\pi k)} = e^{j\Omega_1}e^{j2\pi k} = e^{j\Omega_1}$$

Because of this aliasing, there is a "highest" DT frequency: $\Omega = \pi$.

Check Yourself

Consider 3 CT signals:

$$x_1(t) = \cos(3000t) \quad ; \quad x_2(t) = \cos(4000t) \quad ; \quad x_3(t) = \cos(5000t)$$

Each of these is sampled so that

$$x_1[n] = x_1(nT) \quad ; \quad x_2[n] = x_2(nT) \quad ; \quad x_3[n] = x_3(nT)$$

where $T = 0.001$.

Which list goes from lowest to highest DT frequency?

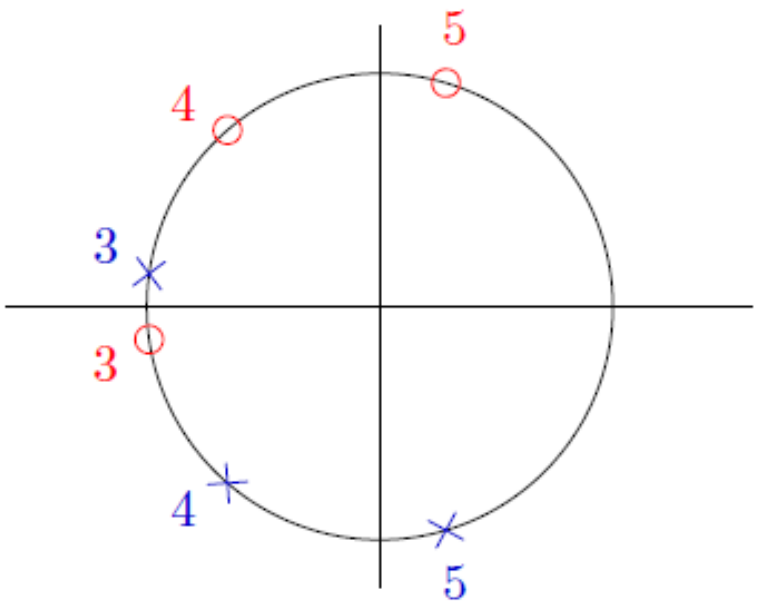
0. $x_1[n]$ $x_2[n]$ $x_3[n]$
1. $x_1[n]$ $x_3[n]$ $x_2[n]$
2. $x_2[n]$ $x_1[n]$ $x_3[n]$
3. $x_2[n]$ $x_3[n]$ $x_1[n]$
4. $x_3[n]$ $x_1[n]$ $x_2[n]$
5. $x_3[n]$ $x_2[n]$ $x_1[n]$

Check Yourself

The discrete signals are

$$x_1[n] = \cos[3n]$$
$$x_2[n] = \cos[4n]$$
$$x_3[n] = \cos[5n]$$

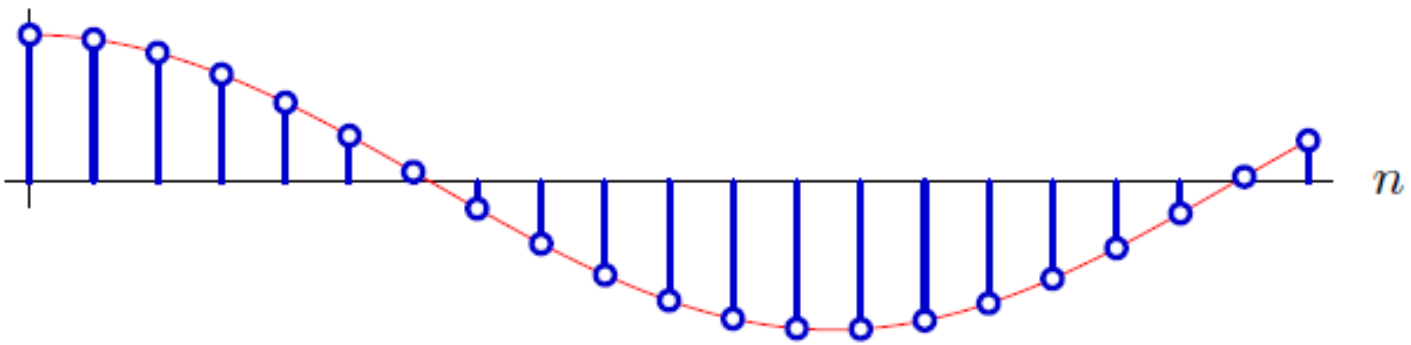
and the corresponding discrete frequencies are $\Omega = 3, 4$ and 5, represented below with $\times$ marking $e^{j\Omega}$ and $\circ$ marking $e^{-j\Omega}$).



Check Yourself

$\Omega = 0.25$

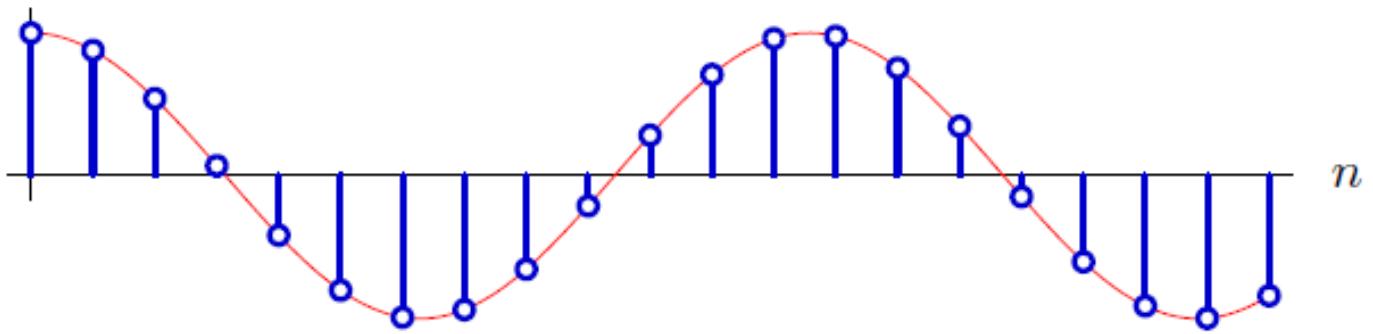
$x[n] = \cos(0.25n)$



Check Yourself

$\Omega = 0.5$

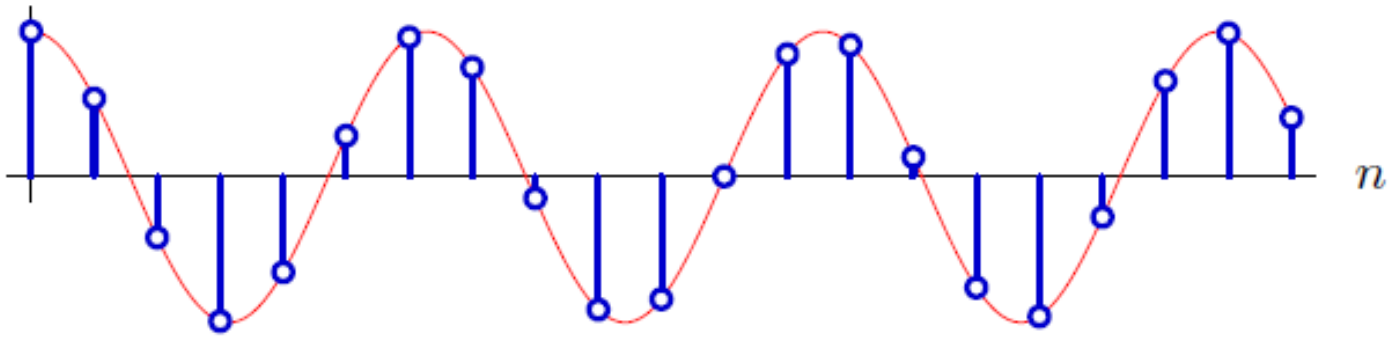
$x[n] = \cos(0.5n)$



Check Yourself

$\Omega = 1$

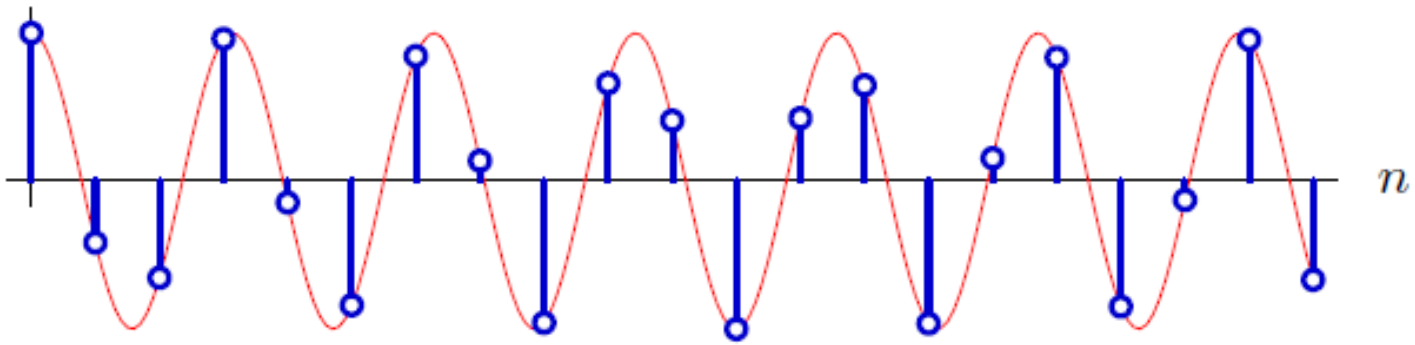
$x[n] = \cos(n)$



Check Yourself

$\Omega = 2$

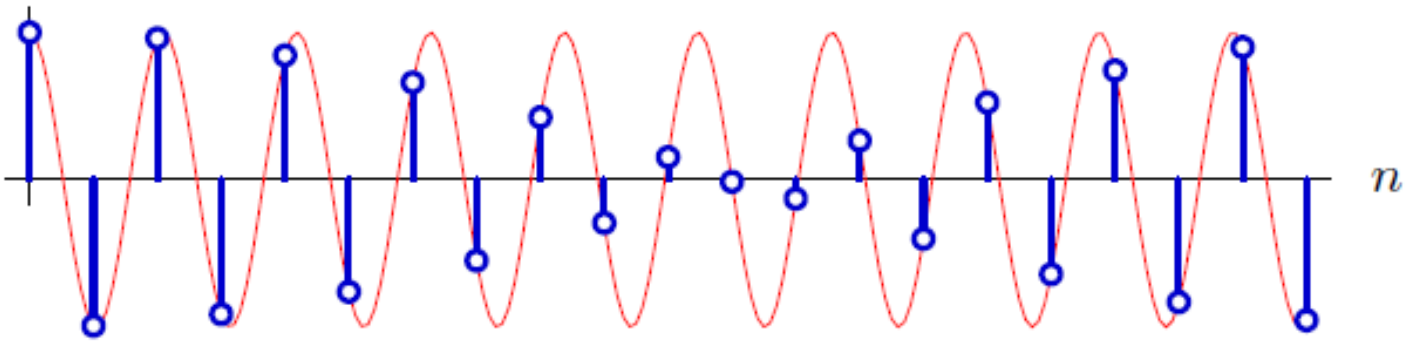
$x[n] = \cos(2n)$



Check Yourself

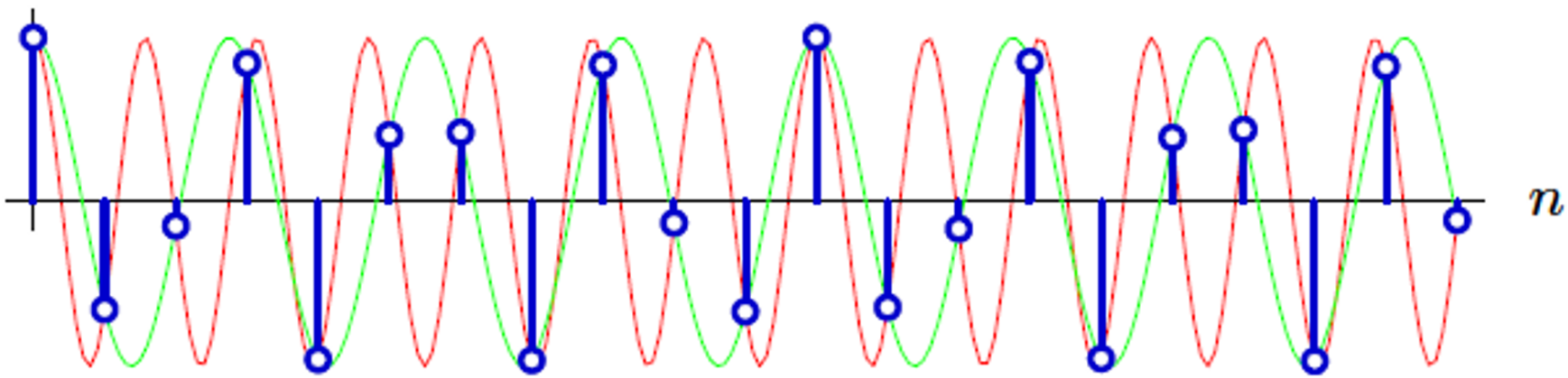
$\Omega = 3$

$x[n] = \cos(3n)$



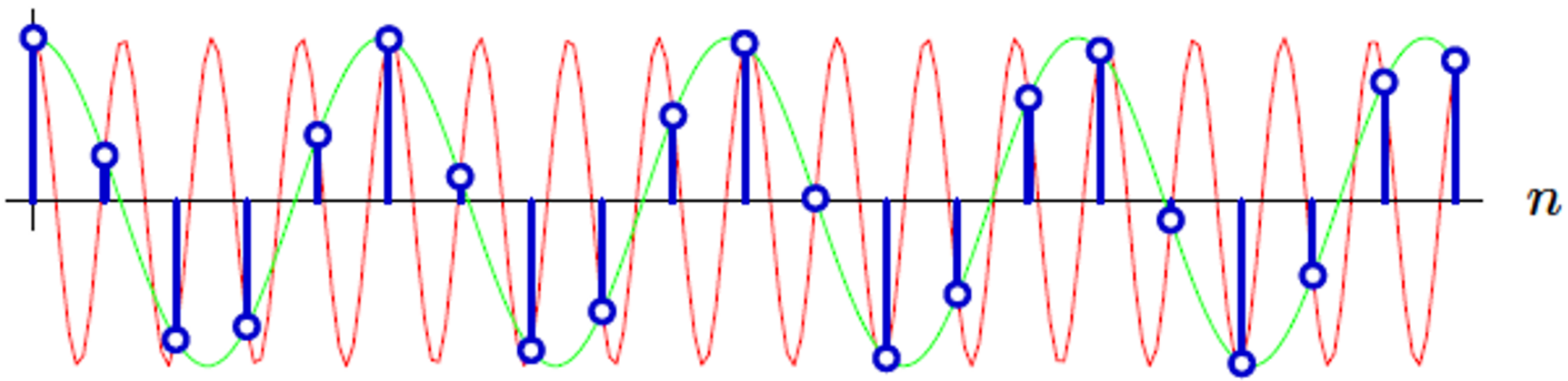
Check Yourself

$$x[n] = \cos(4n) = \cos(2\pi n - 4n) \approx \cos(2.283n)$$



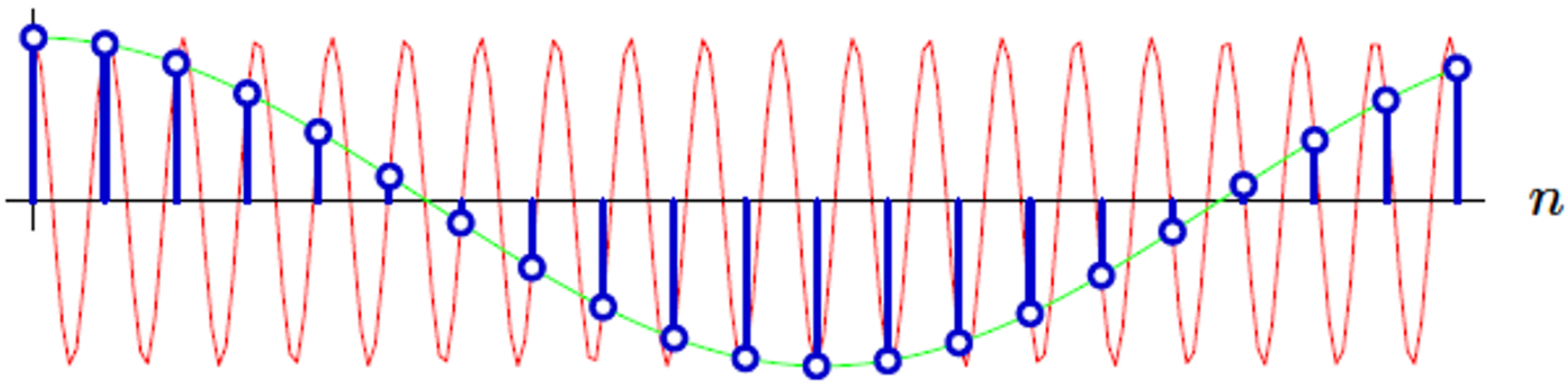
Check Yourself

$$x[n] = \cos(5n) = \cos(2\pi n - 5n) \approx \cos(1.283n)$$



Check Yourself

$$x[n] = \cos(6n) = \cos(2\pi n - 6n) \approx \cos(0.283n)$$

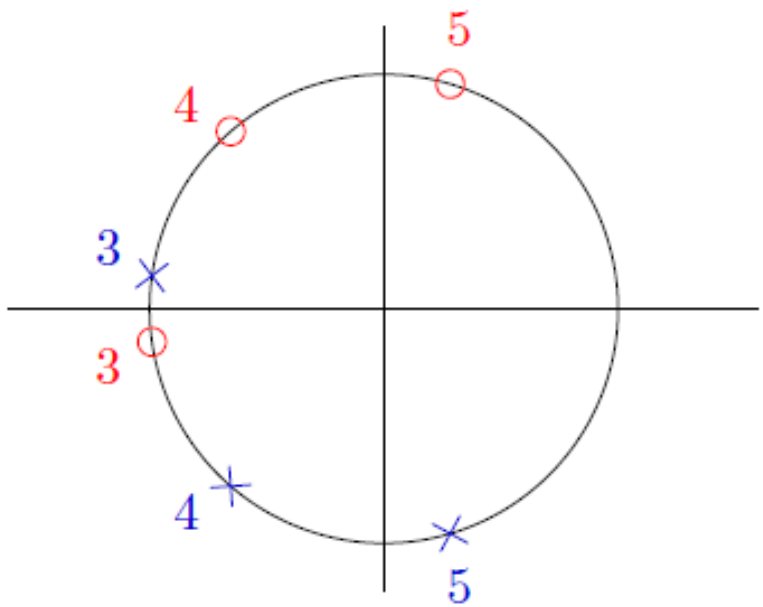


Check Yourself

The discrete signals are

$$x_1[n] = \cos[3n]$$
$$x_2[n] = \cos[4n]$$
$$x_3[n] = \cos[5n]$$

and the corresponding discrete frequencies are $\Omega = 3, 4$ and 5, represented below with $\times$ marking $e^{j\Omega}$ and $\circ$ marking $e^{-j\Omega}$).



Check Yourself

Consider 3 CT signals:

$$x_1(t) = \cos(3000t) \quad ; \quad x_2(t) = \cos(4000t) \quad ; \quad x_3(t) = \cos(5000t)$$

Each of these is sampled so that

$$x_1[n] = x_1(nT) \quad ; \quad x_2[n] = x_2(nT) \quad ; \quad x_3[n] = x_3(nT)$$

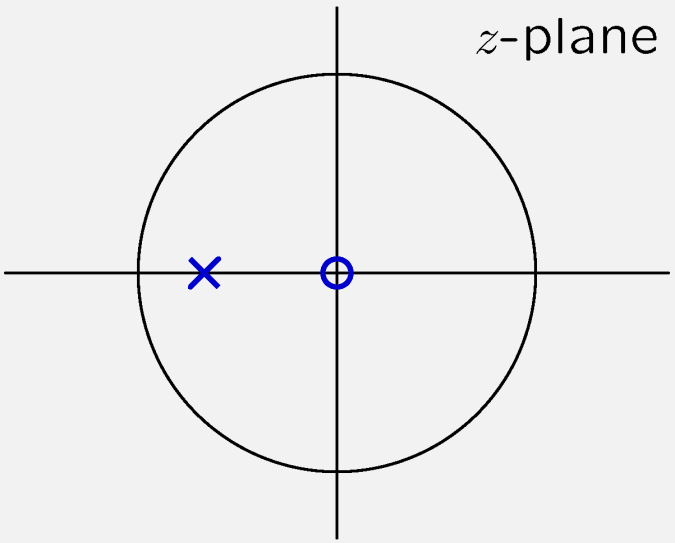
where $T = 0.001$.

Which list goes from lowest to highest DT frequency? 5

0. $x_1[n]$ $x_2[n]$ $x_3[n]$
1. $x_1[n]$ $x_3[n]$ $x_2[n]$
2. $x_2[n]$ $x_1[n]$ $x_3[n]$
3. $x_2[n]$ $x_3[n]$ $x_1[n]$
4. $x_3[n]$ $x_1[n]$ $x_2[n]$
5. $x_3[n]$ $x_2[n]$ $x_1[n]$

Check Yourself

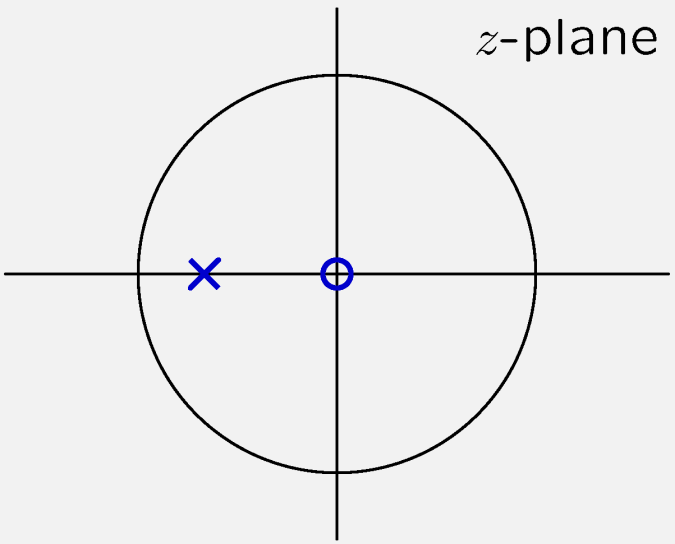
What kind of filtering corresponds to the following?



1. high pass
2. low pass
3. band pass
4. band stop (notch)
5. none of above

Check Yourself

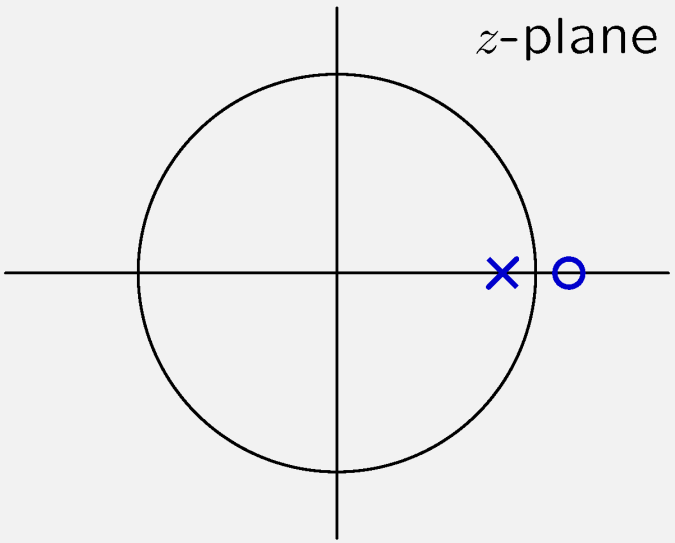
What kind of filtering corresponds to the following? 1



1. high pass
2. low pass
3. band pass
4. band stop (notch)
5. none of above

Check Yourself

A system $H(z) = \dfrac{1 - az}{z - a}$ has the following pole-zero diagram.



Classify this system as one of the following filter types.

1. high pass
2. low pass
3. band pass
4. all pass
5. band stop
0. none of the above

Check Yourself

Classify the system ...

$$H(z) = \frac{1 - az}{z - a}$$

Find the frequency response:

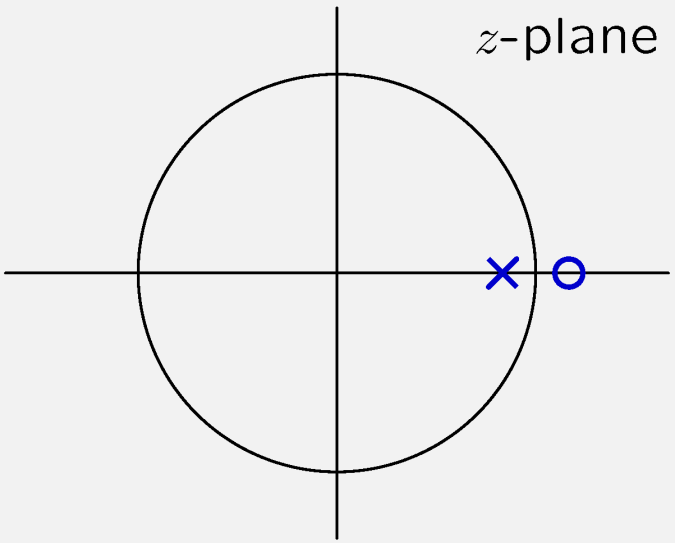
$$H(e^{j\Omega}) = \frac{1 - ae^{j\Omega}}{e^{j\Omega} - a} = e^{j\Omega} \frac{e^{-j\Omega} - a}{e^{j\Omega} - a} \quad \leftarrow \text{complex} \quad \leftarrow \text{conjugates}$$

Because complex conjugates have equal magnitudes, $\left|H(e^{j\Omega})\right| = 1$.

$\rightarrow$ all-pass filter

Check Yourself

A system $H(z) = \dfrac{1 - az}{z - a}$ has the following pole-zero diagram.

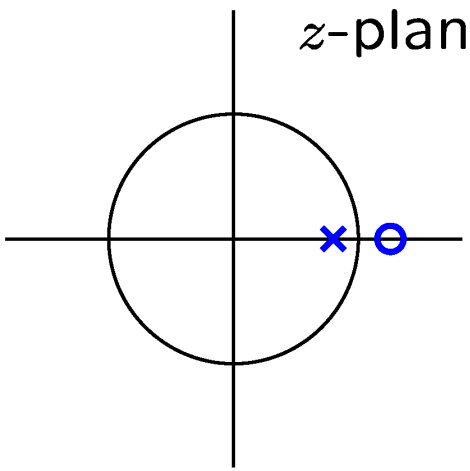


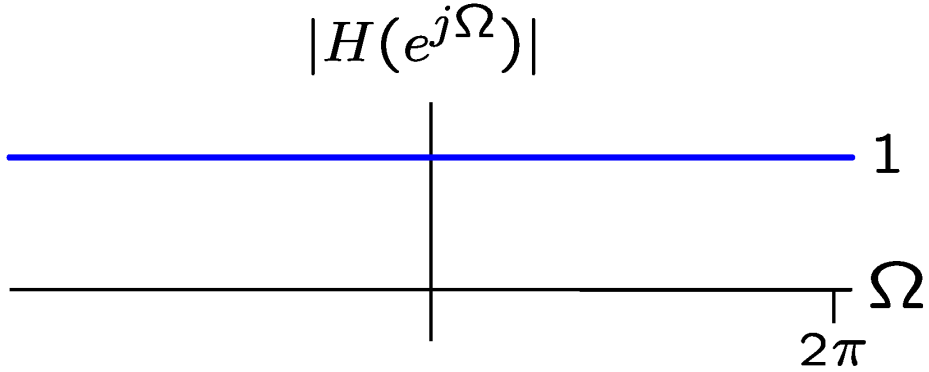
Classify this system as one of the following filter types. 4

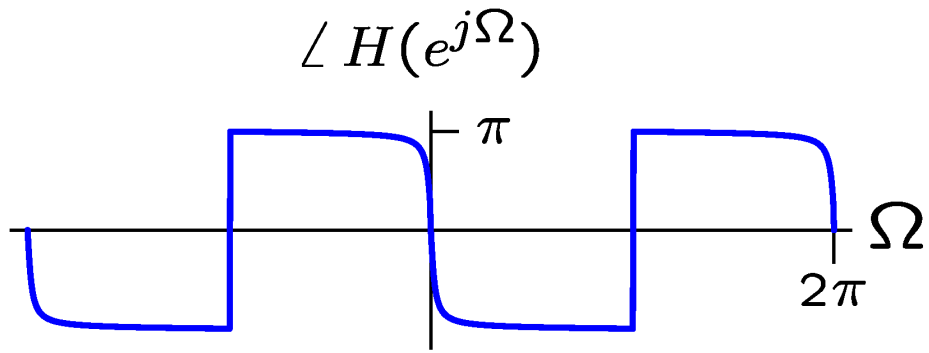
1. high pass
2. low pass
3. band pass
4. all pass
5. band stop
0. none of the above

Effects of Phase

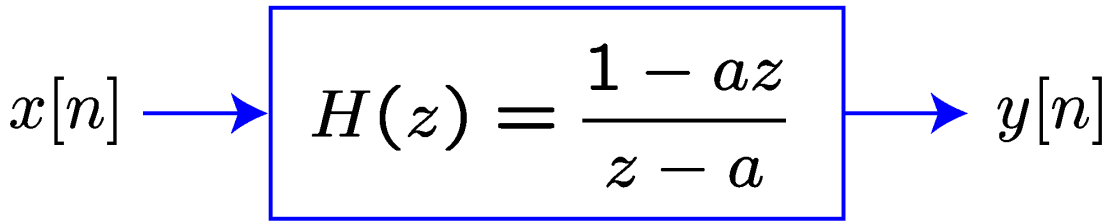
$$x[n] \rightarrow \boxed{H(z) = \frac{1 - az}{z - a}} \rightarrow y[n]$$

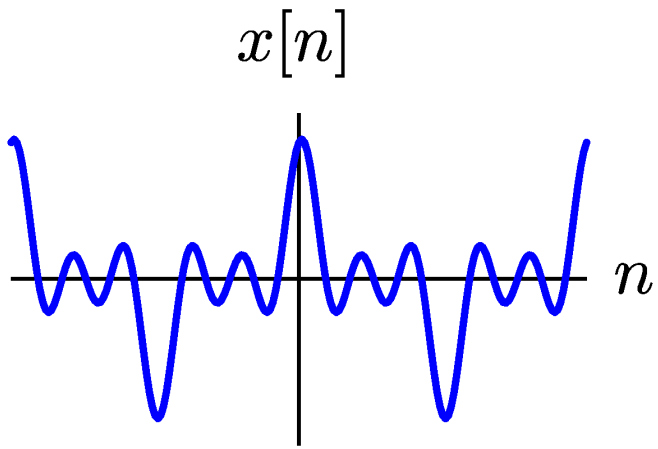


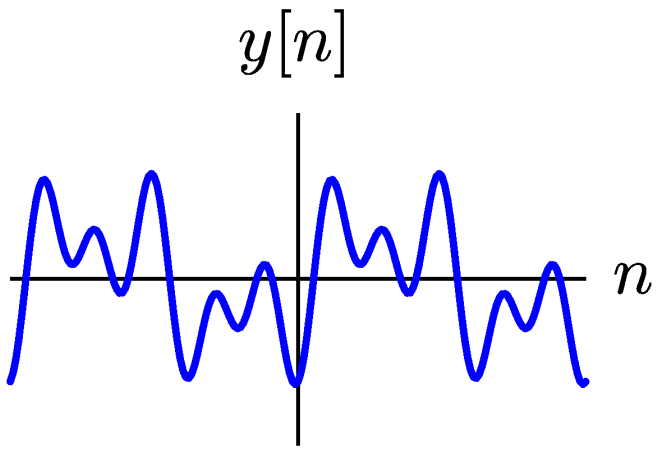




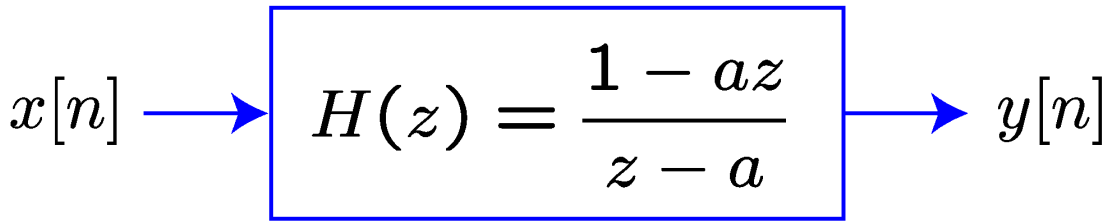
Effects of Phase

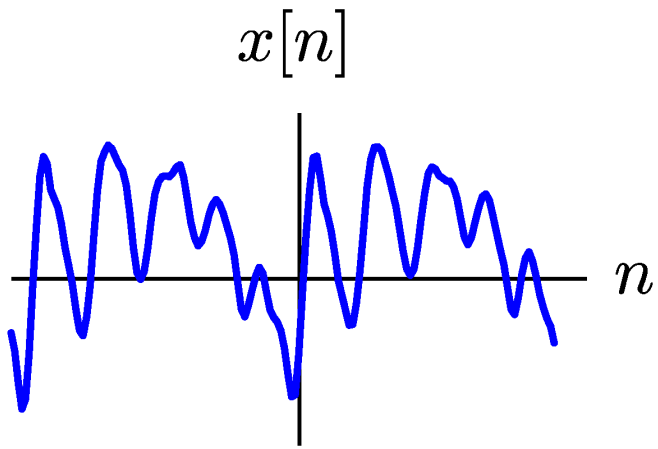


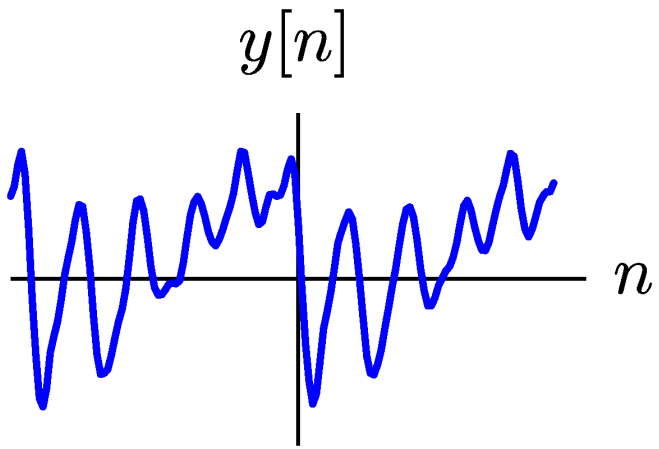




Effects of Phase

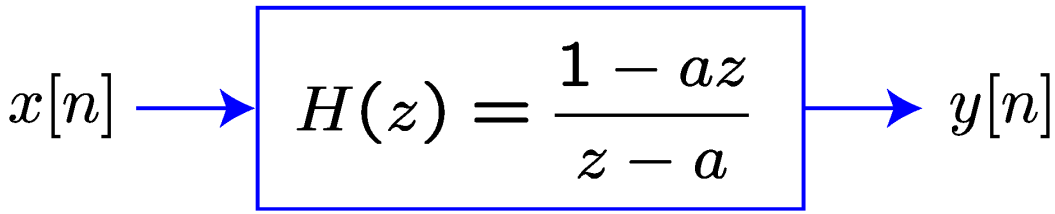


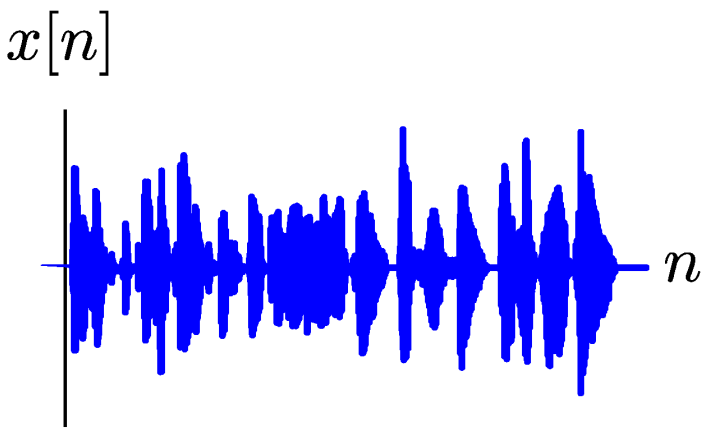


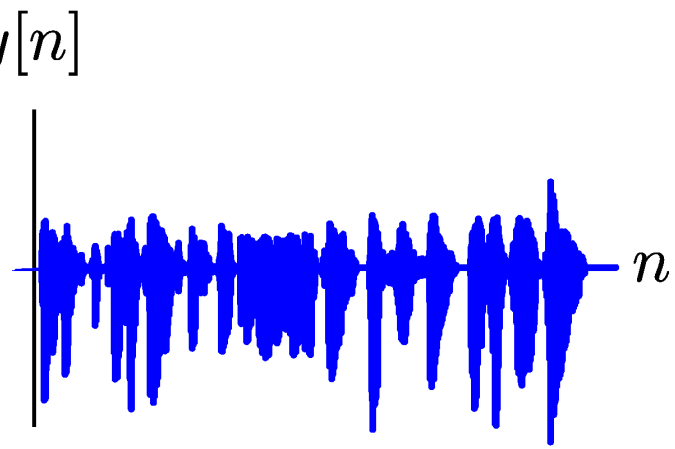


`http://public.research.att.com/~ttsweb/tts/demo.php`

Effects of Phase







`artificial speech synthesized by Robert Donovan`

Assignments

- Reading Assignment: Ch. 3.6-3.11
- Homework 6: Due by May 12, 2025